

Grade 7
Unit 5
Lesson 10 Practice

Name Answer Key
Date _____
Period _____

Denali, in Alaska, is the highest mountain in the United States. Denali is 20,310 feet above sea level. Use this information to answer questions 1-2.

1. Shawna wants to make a scale model of Denali for her geography project. Her teacher suggested using a scale factor of $\frac{700 \text{ ft}}{1 \text{ in}}$. What height should she build the scale model, based on the suggested scale factor?

$$\frac{20310}{700} \approx 29 \text{ inches tall}$$

2. Does Shawna have enough information to build an accurate scale model? Explain.

Answers will vary.

The mountain has multiple dimensions so only having one dimension, the height, isn't enough to create an accurate scale model. You need to have all the dimensions in order to do so.

Mila has a map of Florida showing several major cities hanging in her office. On the map, 1 inch (in.) represents 50 miles (mi). Use this information to answer questions 3-5.

Scale: 1 inch = 50 miles, $\frac{1}{50}$

3. Orlando and West Palm Beach are located 3.4 in. apart on the map. How far apart are the actual cities of Orlando and West Palm Beach?

$$3.4 \text{ in.} \times \frac{50 \text{ mi}}{1 \text{ in.}} = 170 \text{ miles}$$

4. The actual distance between Clermont and Palm Coast is 93 miles. How far apart are these cities on the map?

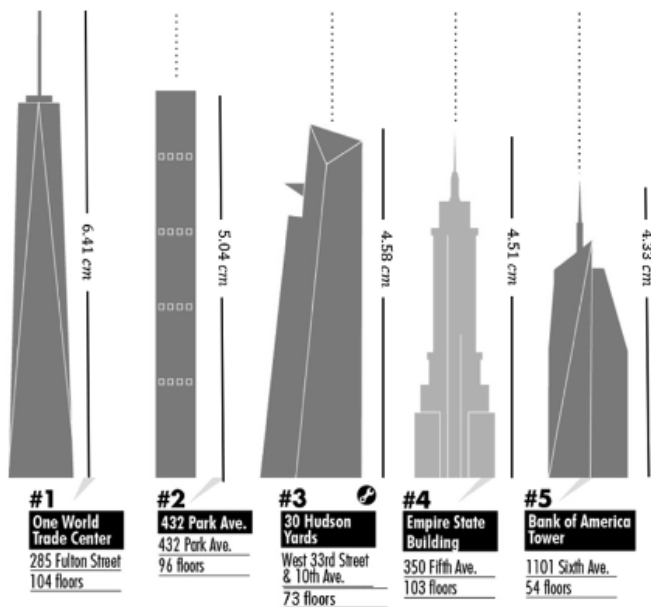
$$93 \text{ mi} \times \frac{1 \text{ in.}}{50 \text{ mi}} = \frac{93}{50} = 1.86 \text{ inches}$$

5. Kelsey is attending the Florida A&M University in Tallahassee. Her top choices for graduate school are Stetson University in Deland or Bethune-Cookman University in Daytona Beach. What information does Kelsey need to determine which university is farthest away from Tallahassee?

Answers will vary.

Kelsey needs to know the distance on the map from Tallahassee to each of her top choices in order to determine the actual distance and which is farthest.

The five tallest buildings in New York City are shown on the scale map below. The heights of the buildings on the figure are also shown in cm. 2 cm represents 554 feet. Use the figure to answer questions 6-8.



6. Based on the given information complete the table.

Building	Actual Height
One World Trade Center	1,775.57 ft
432 Park Ave	1,396.08 ft
30 Hudson Yards	1,268.66 ft
Empire State Building	1,249.27 ft
Bank of America Tower	1,199.41 ft

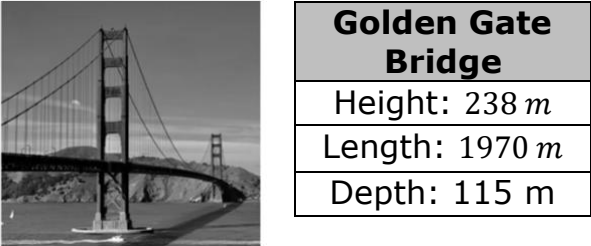
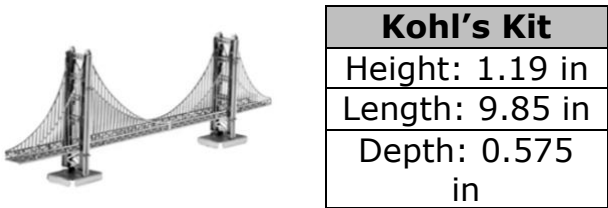
7. Approximately how much taller is One World Trade Center than the Bank of America Tower?

$$1775.57 - 1199.41 = 576.16 \text{ ft}$$

8. Ralph believes that the ratio of the actual height of the Empire State Building to the height of 432 Park Avenue is a little under $\frac{8}{9}$. Is he correct? Explain.

$$\frac{4.51}{5.04} \approx 0.89 \qquad \frac{8}{9} \approx 0.88 \qquad \text{It is a little more than } \frac{8}{9}.$$

The Golden Gate Bridge, in San Francisco California, is one of the most famous and photographed bridges in the world. Rick modified the dimensions of the Kohl’s model kit of the Golden Gate Bridge. Use the figures to answer questions 9-10.



9. Determine whether the Kohl’s metal model an accurate scale model of the actual Gold Gate bridge. Explain.

$$\frac{238}{1.19} = 200 \qquad \frac{1970}{9.85} = 200 \qquad \frac{115}{0.575} = 200 \qquad \text{Yes}$$

10. Ross wants to sell a scale model of the Golden Gate Bridge. What possible scale factor could Ross use to ensure an accurate scale model?

Answers will vary.
In order to make a scale model, the factor from scale to actual needs to be less than one in order to make it doable. A scale factor of less than one ensures that the scale model will be smaller than the actual.